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Annex B

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Operating Systems and Capabilities

Student Outline

Data Systems Administrator Course
M09CVQ1
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Table of Contents

Terminal Learning Objective 2
Enabling Learning Objectives 2
Characteristics of Operating Systems 4
What is an Operating System (OS)? 4
Components of an Operating System (OS) 4
Purpose of an Operating System 7
Types of Operating Systems 9
Types of Operating System Applications 11
Servers 12
How a Server Works 12
Types of Servers 13
Operating System Installation Types 15
Server Operating Systems Edition Types 17
References 19

Terminal Learning Objective

1. Given a command's mission, threat assessment, operational environment, operational plans, virtual environment, and personnel, implement operating systems to establish operating systems in accordance with applicable STIGs and local cybersecurity policies in support of the communications plan. (0671-IMPL-1002)

Enabling Learning Objectives

1. Without the aid of reference, identify in writing the characteristics of operating systems, in accordance with 0671-IMPL-1002.

2. Without the aid of reference, identify in writing the purpose of operating systems, in accordance with 0671-IMPL-1002.

3. Without the aid of reference, identify in writing the types of operating systems, in accordance with 0671-IMPL-1002.

4. Without the aid of reference, identify in writing types of operating system applications, in accordance with 0671-IMPL-1002.

5. Without the aid of reference, identify in writing operating system hardware requirements, in accordance with 0671-IMPL-1002.

6. Without the aid of reference, identify in writing the definition of a server, in accordance with 0671-IMPL-1002.

7. Without the aid of reference, identify in writing the types of servers, in accordance with 0671-IMPL-1002.

8. Without the aid of reference, identify in writing operating system installation types, in accordance with 0671-IMPL-1002.

9. With the aid of reference and given the proper equipment and software, deploy operating systems, in accordance with 0671-IMPL-1002.

10. Without the aid of reference, identify in writing server operating system edition types, in accordance with 0671-IMPL-1002.

Characteristics of Operating Systems

What is an Operating System (OS)?

The term OS or ***operating system*** is a type of software and it works as an interface between the user and computer to perform all the tasks like memory management, file management, input & output handling, security, process management, Job accounting, error detecting, system performance controlling, peripheral devices controlling like printers & disk drives.

The OS is the most significant program in a computer or server because every device works with an operating system (OS) to run all the programs & applications. The main tasks of the computer or server OS are identifying input from the keyboard, sending output to the screen, tracking files, storage drives, peripheral devices controlling, like printers, etc.

Components of an Operating System (OS)

1. ***Kernel***

The kernel in the OS provides the basic level of control on all the computer peripherals. In the operating system, the kernel is an essential component that loads firstly and remains within the main memory. So that memory accessibility can be managed for the programs within the RAM, it creates the programs to get access from the hardware resources. It resets the operating states of the CPU for the best operation at all times.

2. ***Process Execution***

The OS gives an interface between the hardware as well as an application program so that the program can connect through the hardware device by simply following procedures & principles configured into the OS. The program execution mainly includes a process created through an OS kernel that uses memory space as well as different types of other resources.

3. ***Interrupt***

In the operating system, interrupts are essential because they give a reliable technique for the OS to communicate & react to their surroundings. An interrupt is nothing but one kind of signal between a device as well as a computer system otherwise from a program in the computer that requires the OS to leave and decide accurately what to do subsequently. Whenever an interrupt signal is received, then the hardware of the computer puts on hold automatically whatever computer program is running presently, keeps its status & runs a computer program which is connected previously with the interrupt.

4. Memory Management

The functionality of an OS is nothing but memory management which manages main memory & moves processes backward and forward between disk & main memory during implementation. This tracks each & every memory position; until it is assigned to some process otherwise it is open. It verifies how much memory can be allocated to processes and also makes a decision to know which process will obtain memory at what time. Whenever memory is unallocated, then it tracks correspondingly to update the status. Memory management work can be divided into three important groups like memory management of hardware, OS and application memory management.

5. Multitasking

It describes the working of several independent computer programs on a similar computer system. Multitasking in an OS allows an operator to execute one or more computer tasks at a time. Since many computers can perform one or two tasks at a time, usually this can be done with the help of time-sharing, where each program uses the time of a computer to execute.

6. Networking

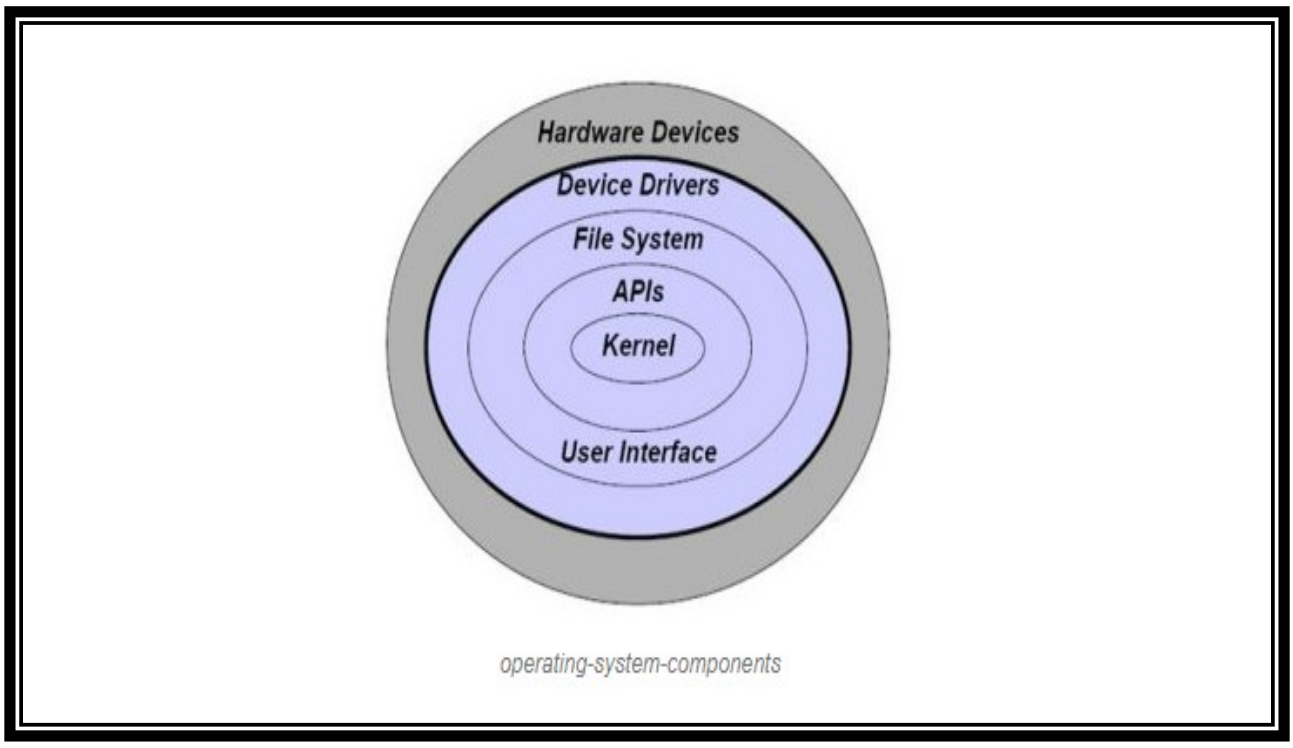
Networking can be defined as when the processor interacts with each other through communication lines. The design of communication-network must consider routing, connection methods, safety, the problems of opinion & security. Presently most of the operating systems maintain different networking techniques, hardware, & applications. This involves that computers that run on different operating systems could be included in a general network to share resources like data, computing, scanners, printers, which uses the connections of either wired otherwise wireless.

7. Security

If a computer has numerous individuals to allow the immediate process of various processes, then the many processes have to be protected from other activities. This system security mainly depends upon a variety of technologies that work effectively. Current operating systems give an entrée to a number of resources, which are obtainable to work the software on the system, and to external devices like networks by means of the kernel. The operating system should be capable of distinguishing between demands which have to be allowed for progressing & others that don't need to be processed. Additionally, to permit or prohibit a security version, a computer system with a high level of protection also provides auditing options. So this will allow monitoring the requests from accessibility to resources

8. User Interface

A GUI or user interface (UI) is the part of an OS that permits an operator to get the information. A user interface based on text displays the text as well as its commands which are typed over a command line with the help of a keyboard.
The OS-based applications mainly provide a specific user interface for efficient communication. The main function of a user interface of an application is to get the inputs from the operator & to provide outputs to the operator. The UI of any application can be classified into two types namely GUI (graphical UI) & CLI (command line user interface).



*The main components of an OS mainly include kernel, API or application program interface, user interface & file system, hardware devices and device drivers.

Purpose of an Operating System

It manages the computer's memory and processes, as well as all of its software and hardware. It also allows individuals to communicate with the computer or server without knowing how to speak the device's language. Without an operating system, a computer or server is useless.

Central Processing Unit

A central processing unit (CPU), also called a central processor, main processor or just processor, is the electronic circuitry that executes instructions comprising a computer program. The CPU performs basic arithmetic, logic, controlling, and input/output (I/O) operations specified by the instructions in the program. This contrasts with external components such as main memory and I/O circuitry, and specialized processors such as graphics processing units (GPUs).

Random Access Memory

Random-access memory (RAM) is a form of computer memory that can be read and changed in any order, typically used to store working data and machine code. A random-access memory device allows data items to be read or written in almost the same amount of time irrespective of the physical location of data inside the memory, in contrast with other direct-access data storage media (such as hard disks, CD-RWs, DVD-RWs and the older magnetic tapes and drum memory), where the time required to read and write data items varies significantly depending on their physical locations on the recording medium, due to mechanical limitations such as media rotation speeds and arm movement.

Hard Disk Drive (HDD)

A hard disk drive (HDD), hard disk, hard drive, or fixed disk is an electro-mechanical data storage device that stores and retrieves digital data using magnetic storage and one or more rigid rapidly rotating platters coated with magnetic material. The platters are paired with magnetic heads, usually arranged on a moving actuator arm, which read and write data to the platter surfaces. Data is accessed in a random-access manner, meaning that individual blocks of data can be stored and retrieved in any order. HDD's are a type of non-volatile storage, retaining stored data even when powered off. Modern HDD's are typically in the form of a small rectangular box.

Solid State Drive (SSD)

A solid-state drive (SSD) is a solid-state storage device that uses integrated circuit assemblies to store data persistently, typically using flash memory, and functioning as secondary storage in the hierarchy of computer storage. It is also sometimes called a solid-state device or a solid-state disk, even though SSD's lack the physical spinning disks and movable read–write heads used in hard disk drives (HDD's) and floppy disks.

Redundant Array of Independent Disks (RAID)

RAID is a data storage virtualization technology that combines multiple physical disk drive components into one or more logical units for the purposes of data redundancy, performance improvement, or both. This was in contrast to the previous concept of highly reliable mainframe disk drives referred to as "single large expensive disk" (SLED).

Hewlett Packard Enterprise

BREAKDOWN OF COMMON RAID LEVELS

RAID LEVEL	METHOD	HARDWARE / SOFTWARE	MINIMUM # OF DISKS	COMMON USAGE	PROS	CONS
JBOD	SPANNING		2	INCREASE CAPACITY	COST-EFFECTIVE STORAGE	NO PERFORMANCE OR SECURITY BENEFITS
0	STRIPING		2	HEAVY READ OPERATIONS	HIGH PERFORMANCE (SPEED)	DATA IS LOST IF ONE DISK FAILS
1	MIRRORING		2	STANDARD APP SERVERS	FAULT TOLERANCE, HIGH READ PERFORMANCE	LAG FOR WRITE OPS, REDUCED STORAGE (BY 1/2)
5	STRIPING & PARITY		3	NORMAL FILE STORAGE & APP SERVERS	SPEED + FAULT TOLERANCE	LAG FOR WRITE OPS, REDUCED STORAGE (BY 1/3)
6	STRIPING & DOUBLE PARITY		4	LARGE FILE STORAGE & APP SERVERS	EXTRA LEVEL OF REDUNDANCY, HIGH READ PERFORMANCE	LOW WRITE PERFORMANCE, REDUCED STORAGE (BY 2/5)
10 (1+0)	STRIPING & MIRRORING		4	HIGHLY UTILIZED DATABASE SERVERS	WRITE PERFORMANCE + STRONG FAULT TOLERANCE	REDUCED STORAGE (1/2), LIMITED SCALABILITY

What Happened to 2-4 and 6-9?

The RAID levels described above are the most common levels used in enterprise scenarios. The levels in between are highly specialized and only make sense in very specific scenarios.

Types of Operating Systems

There are five main types of operating systems. These five OS types are likely what run any data center that can control servers, phones, computers, or other mobile devices like a tablet. Having the knowledge of applications and systems types will help any administrator maintain security and user access, perform routine operations, and much more.

1. Microsoft Windows

The Windows OS has been around since the 1980s and has had several versions and updates (including Windows 95, Windows Vista, Windows 7/8/10/11, etc.) Microsoft Windows is one of the popular operating system types and is preloaded on most new PC hardware. With each new Windows update or release, Microsoft continues to work on improving their users' experience, hardware, and software, making Windows more accessible and easier to use.
Microsoft Windows contains a control panel, a desktop and desktop assistant, disk cleanup, event viewer, and more. Many users prefer Microsoft Windows because they say it's compatible with many other kinds of software. Many kinds of computer programs run best on Microsoft Windows because they're developed by Microsoft.

- ***Workstation***

A **workstation** is a personal computer that is used for high end applications such as graphic design, video editing, CAD, 3-D design, or other CPU and RAM intensive programs. A workstation typically has a top of the line, fast processor, multiple hard drives, and a lot of RAM memory. A workstation may also have special audio, video, or processing cards for special editing work. A workstation is marketed by computer manufacturers to professional users, while the server is more of a utility device.

- ***Server***

A **server** is an application or device that performs service for connected clients as part of client server architecture. It can also be a computer system that has been designated for running a specific server application. The product allows users to share files and services while also providing administrators with control of networks, data storage and applications.

2. Apple macOS

Head-to-head in the competition with Microsoft Windows is Apple's macOS. macOS and Windows are both examples of proprietary operating systems, meaning that the company conceptualized, designed, developed, and now sells their own OS. They're designed and sold by the companies and aren't meant to be tampered with or tweaked by users. Apple and Macintosh computers run on the proprietary macOS and OS X system, the first of which launched 20 years ago. Apple macOS has not made its way onto USMC networks yet however, it could be within the future.

3. Google's Android OS

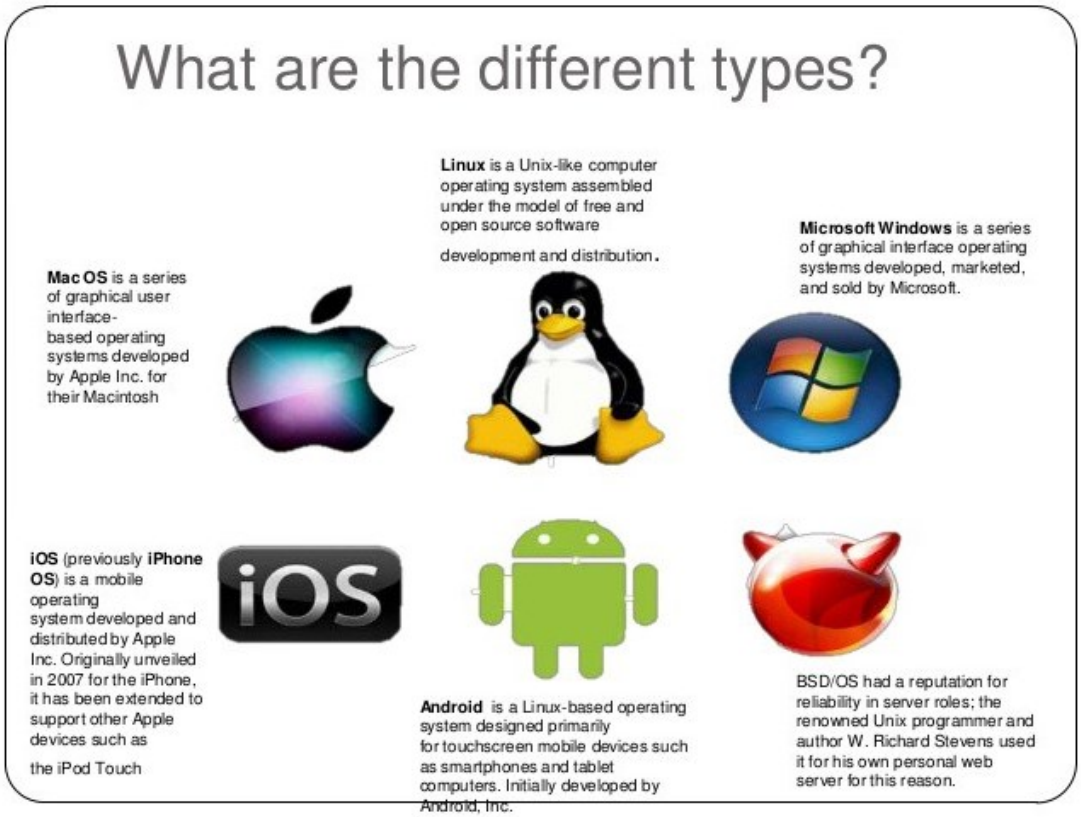
The OS that companies including Google use to run its Android mobile smartphones and tablets is based on Linux distribution and other open source software. Android OS is the primary OS for Google mobile devices like smartphones and tablets. Android has gained increasing popularity since its release as an alternative to Apple's iOS for smartphone users and is continuing to increase in popularity with new updates and exciting features. Android OS is primarily used within the USMC Mobility Infrastructure. This is important because most junior enlisted Marines that are transferred to "Base G6's" are required to provide support for their end users.

4. Apple iOS

Apple's iOS is another mobile operating system used exclusively for iPhones, some of the most popular mobile devices on the market. iOS integrations have regular updates, new expansions to software, and continually are offering new features for users even if they have older devices. Many users appreciate the unique user interface with touch gestures, and the ease of use that iOS offers. This operating system also allows other Apple devices to connect, giving users easy connections to other devices or people. As with Android OS, iPhones are now operating on the Marine Corps Enterprise Network and fall under the Mobility Infrastructure.

5. Linux Operating System.

Linux is different from Windows and Apple in that it's not a proprietary software, but rather a family of open source systems. In other words, anyone can modify and distribute it. Linux may be the least known on this list, but it's free and available in many different open source versions. Linux is popular because of its ease of customization and offers a variety of options to those who understand how to use it. Linux systems that are used on Marine Corps networks are as follows: Cisco Call Manger, ACAS, and even some remote administration servers.



Types of Operating Systems

Types of Operating System Applications

Is an operating system an application program?

The operating system itself can be considered application software when performing simple calculating, measuring, rendering, and word processing tasks not used to control hardware via command-line interface or graphical user interface

The main difference between operating system and application software is that an operating system is a system software that works as the interface between the user and the hardware while the application software is a program that performs a specific task.

Application software, or app for short, is software that performs specific tasks for an end-user. For example, Microsoft Word or Excel are application software, as are common web browsers such as Firefox or Google Chrome.

Servers

Definition of a Server

A server is a computer program or device that provides a service to another computer program and its user, also known as the client. In a data center, the physical computer that a server program runs on is also frequently referred to as a server (The ESXi host). *This is done by server virtualization, previously taught in Annex B.

In the client/server programming model (typically any USMC architecture), a server program awaits and fulfills requests from clients that are on the network. Clients can be other users making request or other servers on the network that are trying to process information from one another.

A server may be designed to do a single task, such as a mail server, which accepts and stores email and then provides it to a requesting client. Servers may also perform several tasks, such as a file and print server, which both stores files and accepts print jobs from clients and then sends them on to a network-attached printer.

How a Server Works

To function as a server, a device must be configured to listen to requests from clients on a network connection. This functionality can exist as part of the operating system as an installed application, role, or a combination of the two.

For example, Microsoft's Windows Server operating system provides the functionality to listen to and respond to client requests. Additionally installed roles or services increase which kinds of client requests the server can respond to.

When a client requires data or functionality from a server, it sends a request over the network. The server receives this request and responds with the appropriate information. This is the request and response model of client-server networking, also known as the call and response model.

A server will often perform numerous additional tasks as part of a single request and response, including verifying the identity of the requestor, ensuring that the client has permission to access the data or resources requested, and properly formatting or returning the required response in an expected way.

Types of Servers

1. ***Application Server***

This is an environment where different applications can run irrespective of what they are or what functions they will perform. It is utilized in developing, processing and running web-based applications. Some of the examples are PHP, Java, and .NET Framework. Application, they provide the advantage of processing high-end jobs and centralized storage of data.

2. ***Cloud Server***

They are highly distributed, massive storage and high-end processing power servers. These types are mainly used by "Cloud Computing" to serve multiple request at one time.

What is Cloud Computing?

It is an environment that is accessed through an application (google drive) on computers, smartphones, tablets or through the website that access the cloud through the browser.

Examples: Amazon Web Service, Google Docs, Window Azure.*USMC is starting to make the transition to the cloud so make sure to understand the difference between on premise and off-premise services. *Microsoft TEAMS is "cloud" based.

3. ***Database Server***

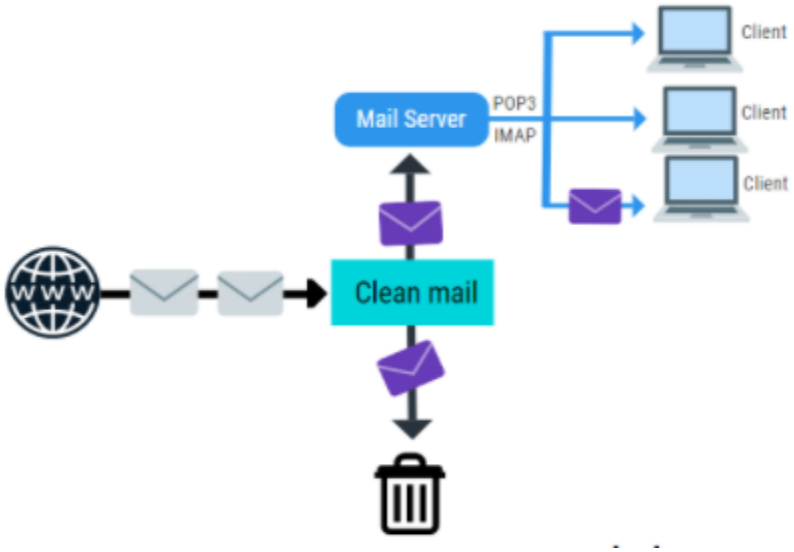
They provide access and retrieving of data from a database. We can access the database by "frontend", which runs locally user machine and "backend" termed as running on the database itself accessed by any remote device or shell. Any information in the database is retrieved by firing query then it is outputted to the user requesting the data. The big organization utilizes a database for storage and users can access the data by executing a query using any query language.

Example: SCCM and SQL Servers contain massive db's that are administered by 0671's.

4. ***Mail Server***

It is a centralized or remote computer that holds e-mail messages for the client on a particular network. It is the same as to the post office, where mails are stored and sorted before dispatching it to the final destination. Whenever the user requests or sends any e-mail through the client machine, they need to contact the mail server, to deliver mail to receiving the client's computer.

Examples: Simple Mail Transfer Protocol (SMTP), Post office Protocol 3(POP3). USMC Networks currently use Microsoft Exchange 2016, 2019 and Exchange O365



5. ***Domain Name Service Server***

It is popularly known by Domain Name System which receives a request having a domain name hostname and responds back corresponding IP address. DNS uses port 53 for providing the service.

6. ***Proxy Server***

It is a computer that's part of the gateway server or another computer which separates a local network from worldwide networks. It caches all the pages accessed through the network which will allow users who want to visit that same page, which will load the load faster and reduce the network's bandwidth. When a page is accessed without proxy server's access the web page using its own IP address caches that page and forwards it to the respective user.



Operating System Installation Types

The three most common installation methods of Windows are?

DVD Boot/.ISO installation, Distribution share installation, image based installation.

What's an ISO?

Any file that ends in .iso is an exact copy of a disk. It's a virtual copy of a CD or DVD, with the same file structure and the same data. ISO copies are referred to as "images" of the original. The acronym ISO comes from the name of the organization responsible for creating industry standards—International Organization for Standardization. The ISO is a perfect copy of what would be on an original Windows CD or DVD.

When administrators download the ISO file, they must copy it onto a USB or DVD. When there're ready to install Windows, "mount" the USB drive or DVD with the ISO file on it and then run Setup.exe from the root folder on the drive.

This allows them to install Windows onto their virtual machine without having to first run an existing operating system. If they can change the boot order of drives in their computer's BIOS, they can run the Windows installation directly from the virtual machine USB drive or DVD when they first turn on the computer. Please see the documentation for their specific computer for information about how to change the BIOS boot order of drives.

Distribution Share Installation

A distribution share is an optional set of folders that contain custom scripts, images, branding, applications, drivers, and other files. These files can be copied to Windows® during installation through an answer file (Unattend.xml).

During installation, Windows connects to the path of the server share by using the credentials that you specify in an answer file. Only the files that you specify in the answer file are copied to the Windows installation.

If you are installing Windows in an environment that does not have a network share or server share, you can copy the necessary files from the distribution share to a configuration set. A configuration set is a subset of the files in a distribution share. You can copy a configuration set to external storage, such as a USB flash drive or an external hard disk, to use during installation. *In USMC networks, this method is the least likely however, it is a possibility that can be depended on if needed.

Image Based Installation

After installing an operating system on reference system and setting it up with specific settings and configuring other applications they would leverage an imaging solution to capture an image based on a sector-by-sector copy of the reference computer. A good example of this is Microsoft's System Center Configuration Manager (SCCM).

What is SCCM? How Does SCCM Work

Microsoft's SCCM (System Center Configuration Manager) is a paid lifecycle management solution from Microsoft that keeps track of a network's inventory, assists in application installation, and deploys updates and security patches across a network. While SCCM uses Microsoft's WSUS patching system to check for and install updates, it gives users additional patch management control over when and how patches are applied, and includes many more features which make it an attractive option for large enterprise networks. However, Microsoft SCCM presents several challenges for organizations looking for one solution to provide patch management across all devices, operating systems, and third party applications, so it is important to evaluate the pros and cons of patching with SCCM.

There are two types of installation

*Guided Install: Install an operating system and device drivers in an unattended mode.
*Manual Install: Install an operating system and device drivers manually.

Windows Setup can perform both clean and upgrade installations. However, it does not perform computer-to-computer migrations. Instead, administrators must use Windows Easy Transfer, the User State Migration Tool (USMT), or another migration tool to move data from a previous installation to the new operating system.

- **Custom installations.** Windows Setup can perform a custom installation, also known as a clean installation, which will save previous Windows installation but does not migrate settings. The previous Windows installation will not boot after a clean installation.

- **Upgrade installations.** Windows Setup can perform an installation that retains your settings and preferences while upgrading your operating system.

Server Operating Systems Edition Types

Introduction

Windows Server 2019 is an operating system created by Microsoft and created to run on servers. Servers are specialized computers that usually operate within the client-server network; servers handle requests from the clients on the network. The use of a server operating system can allow other applications to run on the server's hardware, and this can help fill different server job functions, such as a mail server or print server.
Microsoft Windows Server 2019 is based on the popular Windows Server 2016, and aims to reduce business risk and enhance security through the multiple layers of security and protection embedded in the operating system. One of the difference from the previous versions is the addition of hyper-converged infrastructure (HCI) and Hybrid capabilities with Azure (which provides a number of cloud services like virtual computing, analytics, storage and networking).

Microsoft Windows Server 2019 is available in 2 editions, ***standard and datacenter***. Choosing the right edition is extremely important so administrators don't end up buying something which is either incapable of fulfilling their organizations wants, or has unnecessary features which will add to their costs.

Windows Server 2019 Standard

Windows Server Standard 2019 is suitable for small and medium organizations. This edition allows administrators to use up to 2 Virtual Machines with 1 Hyper-V host for each license. If administrators want to add 2 more virtual machines, they have to license all the cores in the server again.

Windows Server 2019 Standard enables the ability to modernize apps quickly, and download them faster, while enhancing the support system. The Standard edition also lacks Host Guardian Hyper-V Support and thus does not ensure that the host knows about the Hyper-V hosts. Also, it does not allow the ability to manage the keys required to start shielded virtual machines.

This Windows Server 2019 Standard edition is also equipped with a Storage Replication; limited to 1 partnership and 1 resource group. It allows the administrators to replicate a single volume of data up to 2 TB.

Administrators can activate this edition as a guest only if the virtual machines are hosted on the Datacenter edition. Also, the Standard edition lacks features like Software-defined Networking (which is responsible for enabling the ability to manage the network efficiently and monitor and improve its performance) and Storage Spaces Direct (responsible for creating a highly-available software-defined storage at a lower cost through the use of industry-standard servers along with local-attached drives).

Windows Server 2019 Datacenter

The Windows Server Datacenter edition offers the widest range of capabilities and features without a large number of licensing limitations. It allows administrators to use an unlimited number of virtual machines, with 1 Hyper-V host for each license.

This edition is suitable for organizations demanding advances storage and virtualization, and who have more demanding Information technology needs.

The Datacenter edition is equipped with unique product features. One of them is the Host Guardian Hyper-V Support. This also ensures that the enterprise or hosting company is aware of the Hyper-V hosts.

The Windows Server 2019 Datacenter edition also has Storage Replica with no limits on the amount of data or number of volumes. This allows administrators to replicate or back up all of their data for disaster recovery, and ensure smooth functioning of their website.

Administrators can activate this edition either as a guest or as a host. This edition also has the feature of Software-defined networking, which allows them to program the network in a centrally-controlled manner. It also enables the ability to monitor and improve the network holistically, notwithstanding the network technology.

Another feature is Storage Spaces Direct, which gives incomparable performance and efficiency when it comes to storage of data. It uses industry-standard, highly-available servers to store the data in local-attached drives. All this comes at a price cheaper than the traditional NAS or SAN arrays.

Another advantage of the Datacenter edition is that administrators can virtualize an unlimited number of Windows Server instances within the physical Windows Server machine.

Features available generally	Windows Server 2019 Standard	Windows Server 2019 Datacenter
Best Practices Analyzer	Yes	Yes
Direct Access	Yes	Yes
Dynamic Memory (in virtualization)	Yes	Yes
Hot Add/Replace RAM	Yes	Yes
Microsoft Management Console	Yes	Yes
Minimal Server Interface	Yes	Yes
Network Load Balancing	Yes	Yes
Windows PowerShell	Yes	Yes
Server Core installation option	Yes	Yes
Server Manager	Yes	Yes
SMB Direct and SMB over RDMA	Yes	Yes
Software-defined Networking	No	**Yes**
Storage Migration Service	Yes	Yes
Storage Replica	Yes, (1 partnership and 1 resource group with a single 2TB volume)	Yes, **unlimited**
Storage Spaces	Yes	Yes
Storage Spaces Direct	No	**Yes**
Volume Activation Services	Yes	Yes
VSS (Volume Shadow Copy Service) integration	Yes	Yes
Windows Server Update Services	Yes	Yes
Windows System Resource Manager	Yes	Yes
Server license logging	Yes	Yes
Inherited activation	As guest if hosted on Datacenter	**Can be a host or a guest**
Work Folders	Yes	Yes

** Details from Microsoft website at https://docs.microsoft.com/en-us/windows-server/get-started-19/editions-comparison-19

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